

Tucker factorization and closure under mode products

Proof of Conjecture 00000002506

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We prove the stated algebraic structure for arbitrary finite-order coordinate tensors. Let D be a finite set of modes; for each $i \in D$, let I_i, J_i, L_i be finite index sets. Over any commutative semiring K (in particular $\mathbb{R}$ or $\mathbb{C}$), a core is an actual array $C : \prod_i J_i \rightarrow K$ and the factors are rectangular matrices $U_i : I_i \times J_i \rightarrow K$. Define their Tucker contraction by

$$[\mathcal{T}_U C](x) = \sum_{y \in \prod_i J_i} C(y) \prod_{i \in D} U_i(x_i, y_i).$$

This is exactly a core times one factor matrix in every mode. A product in a single mode is obtained by taking its designated factor matrix and identity matrices in all other modes.

For arbitrary further factors $V_i : L_i \times I_i \rightarrow K$, finite distributivity gives the full closure identity

$$\mathcal{T}_V(\mathcal{T}_U C) = \mathcal{T}_W C, \quad W_i = V_i U_i.$$

Indeed its left-hand side at z is

$$\begin{aligned} \sum_x \sum_y C(y) \prod_i U_i(x_i, y_i) \prod_i V_i(z_i, x_i) &= \sum_y C(y) \sum_x \prod_i [V_i(z_i, x_i) U_i(x_i, y_i)] \\ &= \sum_y C(y) \prod_i \left[\sum_{x_i \in I_i} V_i(z_i, x_i) U_i(x_i, y_i) \right] \\ &= [\mathcal{T}_W C](z). \end{aligned}$$

All sums are finite, so no convergence assumption is required. Thus every given Tucker representation remains a Tucker representation after mode products, with the same core and updated factor matrices. In one mode, successive factors A then B combine as BA . In distinct modes, they commute, since in every mode at most one of the two factors is nonidentity; both orders give the same updated tuple.

Existence of a Tucker representation without additional compression requirements is immediate but also explicit: take $C = T$ and every factor to be its identity matrix. Then $\prod_i \delta_{x_i, y_i}$ is one precisely when $y = x$ and zero otherwise, hence $\mathcal{T}_{\text{id}} T = T$. This covers every tensor, including zero tensors, empty mode sets and empty index sets. The source states algebraic factorization and closure; it does not require a smallest core, optimal multilinear rank or orthogonal factors, and no such stronger conclusion is asserted here.

Formal verification. Lean uses genuine dependent coordinate arrays and actual rectangular matrices for arbitrary finite mode/index types. It proves the finite contraction composition identity, the identity kernel and factorization, universal existence and closure of Tucker representations, same-mode composition and distinct-mode commutation. The seven final theorem audits use only standard Lean axioms; no decomposition or contraction certificate is assumed.